

Analog Front-End (AFE) for Mixed-Signal AQI Meter

1 Executive Summary

This report details the design, simulation, and validation of an Analog Front-End constructed in LTspice. The system interfaces with a PM2.5 Optical Dust Sensor and a CO2 Metal Oxide Semiconductor (MOx) sensor. The core objective was to extract raw, noisy, non-linear physical signals, condition them through analog amplification and active filtering, and strictly protect the downstream 3.3V digital microcontroller (ADC) from overvoltage faults.

2 Sensor Transduction & Environmental Modeling

To validate the circuit against real-world physics, static inputs were abandoned in favor of dynamic mathematical behavioral modeling representing a 24-hour office environment.

2.1 PM2.5 Optical Sensor (Sharp GP2Y1010AU0F)

- **Operating Principle:** Linear light scattering. Microsecond light pulses are converted to nano-ampere currents. Highly vulnerable to 60Hz Electromagnetic Interference (EMI) from AC power lines.
- **LTspice Implementation:** A Behavioral Current Source (bi) replacing a static photodiode.
- **Input Formula:** $I = (10n * V(DUST_LEVEL)) + 50n * \sin(120 * \pi * time)$
- **Expected Output:** Converts a massive $140 \mu g/m^3$ dust spike (representing janitorial sweeping) into a measurable baseline current while deliberately injecting a 60Hz AC sine wave to test downstream filters.

2.2 CO2 MOx Sensor (MQ-135 / MG-811)

- **Operating Principle:** Logarithmic chemical surface reaction. Vulnerable to slow thermal/chemical baseline drift due to internal 300°C heaters.
- **LTspice Implementation:** A dynamic, math-driven current source embedded inside a physical Wheatstone bridge to prevent “silent fails” from static resistor limits.
- **Input Formula:** $I = V(I_{n+}) / (10k / (1 + \log_{10}(V(CO2_LEVEL) / 400)))$
- **Expected Output:** The resistance naturally sits at **10 kΩ** at a 400ppm fresh-air baseline (perfectly balancing the bridge to 0V differential). As CO2 rises to 1200ppm, the resistance drops to roughly **6.77 kΩ**, creating a non-linear differential voltage unbalance.

3 Signal Conditioning & Amplification

This stage translates the raw sensor physics into clean, readable voltages scaled perfectly for a 3.3V ADC.

3.1 PM2.5 Transimpedance Amplifier (TIA) & Filtering

Component	Value/Part	Justification
Op-Amp	LT1056	Replaced generic symbols to prevent “unknown subcircuit” errors. Provides real-world bandwidth and stability for high-speed current-to-voltage conversion.
Feedback Resistor	1.5 M Ω	Scales the tiny nano-ampere photodiode currents into the Volt range.
Compensation Cap	10 pF / 21 nF	Cancels the parasitic capacitance of the photodiode to ensure phase margin stability and prevent high-frequency oscillation.
MFB Active Filter	4th-Order RC	Designed to establish a strict low-pass cutoff ($f_c = 5\text{Hz}$) to crush the injected 60Hz EMI sine wave into a flat line.

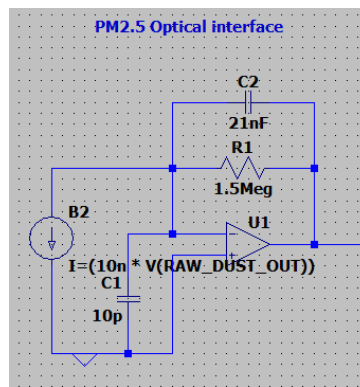


Figure 1: PM2.5 Optical Sensor Module

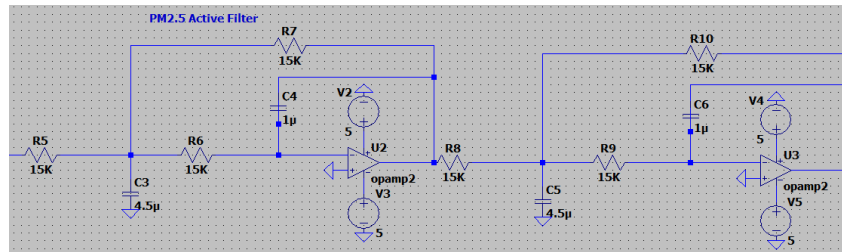


Figure 2: MFB Active Filters

3.2 CO2 Instrumentation Amplifier (LT1167)

Component	Value	Justification & Iterative Tuning
Bridge Resistors	10 k Ω	R_2, R_3, R_4 configured to match the baseline sensor resistance.
Gain Resistor (R_G)	1.5 k Ω	<i>Tuning History:</i> Initially 1 k Ω (caused power rail clipping at 9.2V). Then 10 k Ω (caused op-amp headroom clipping at 4.2V). Final optimal value calculated to provide a Gain of ~ 33 . This stretches the tiny 75mV bridge unbalance into a full 2.5V swing for maximum ADC resolution.
REF Pin Bias	0.5V	Lifted the pin from Ground. Op-amps have a physical “floor” (approx 150mV). Biasing to 0.5V prevents the baseline from crashing into the floor, giving the signal room to breathe.

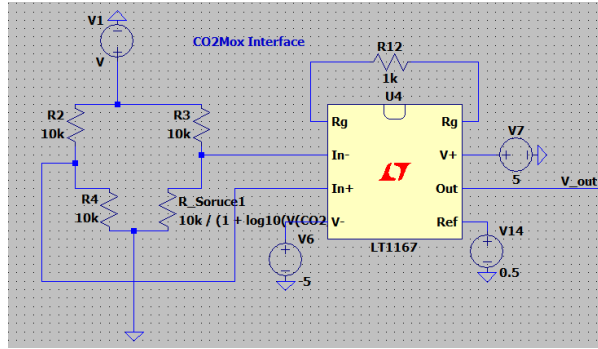


Figure 3: CO2 Mox Interface

4 ADC Interface & Overvoltage Protection

Microcontrollers operate on strict 3.3V limits. This stage ensures perfect quantization mapping and absolute physical safety.

4.1 Precision Voltage Reference

- **Component:** LT1009 or LT1460 (2.5V Precision Reference).
- **Bias Resistor:** 500 Ω (connected to +5V to provide $\sim 5\text{mA}$ activation current if using a shunt type like LM336).
- **Derivation Divider:** 40 k Ω / 10 k Ω . Drops the 2.5V reference down to exactly **0.5V** to feed the LT1167 REF pin cleanly.
- **Expected Output:** A rock-solid 2.5V line to feed the external VREF pin of the microcontroller, isolating the ADC from noisy 3.3V digital power supply fluctuations.

4.2 The Protection Chain (Sequential Flow)

1. **Unity Gain Buffer (LT1056 / UniversalOpamp2):** Isolates the sensitive filter stages from the current draw of the protection diodes.
2. **Current-Limiting Resistor (150 Ω):** Prevents infinite fault currents. If the op-amp shorts to 5V, this resistor chokes the current to a safe $\sim 10\text{mA}$ so the diodes do not melt.
3. **Schottky Diode Clamp (2x 1N5819):**
 - *Top Diode:* Anode to signal, Cathode to an active **3.3V** source. Clamps any signal above 3.6V.
 - *Bottom Diode:* Anode to Ground, Cathode to signal. Clamps any signal below -0.3V.
4. **Charge Bucket Filter (33 Ω + 1 nF):** An anti-kickback RC reservoir sitting immediately at the ADC pin. Instantly dumps electrons into the microcontroller's internal Sample-and-Hold capacitor during sampling, preventing the op-amp from destabilizing.

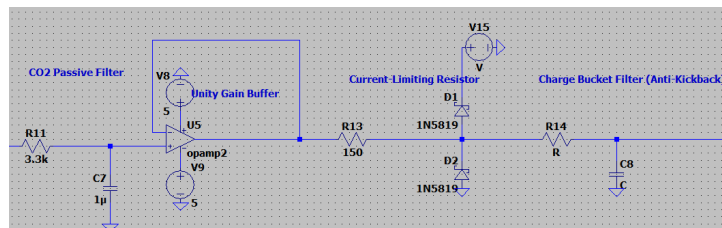


Figure 4: ADC Interfacing

5 Simulation & Validation Strategy

To prove the mathematical theories, three specific SPICE analyses were utilized.

5.1 Transient Analysis (.tran 24)

- **Purpose:** Time-domain testing of the 24-hour environmental PWL text files.
- **Expected Output:** Proved that the 0.5V baseline holds steady during the “empty office” hours. Validated that the final CO2 signal peaks at exactly **3.0V** during the 1200ppm afternoon rush without flatlining (no clipping). Demonstrated the fast rise time of the dust sensor during the 140 $\mu\text{g}/\text{m}^3$ sweeping event.

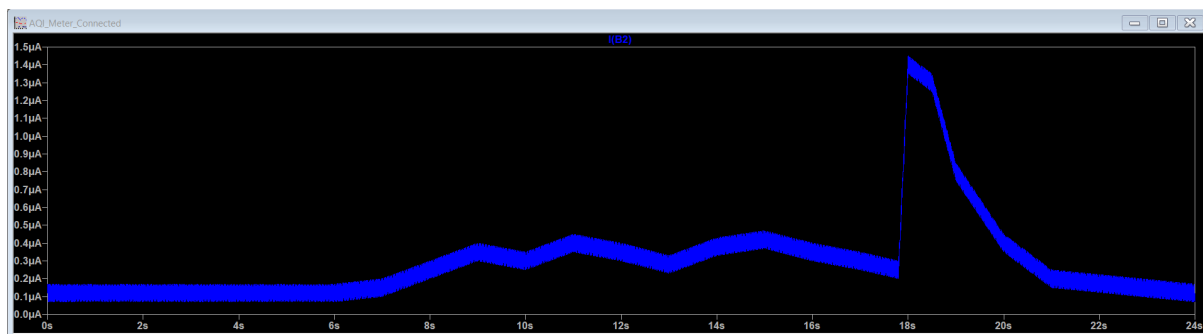


Figure 5: Input Current that the Sensor sees (including the errors)

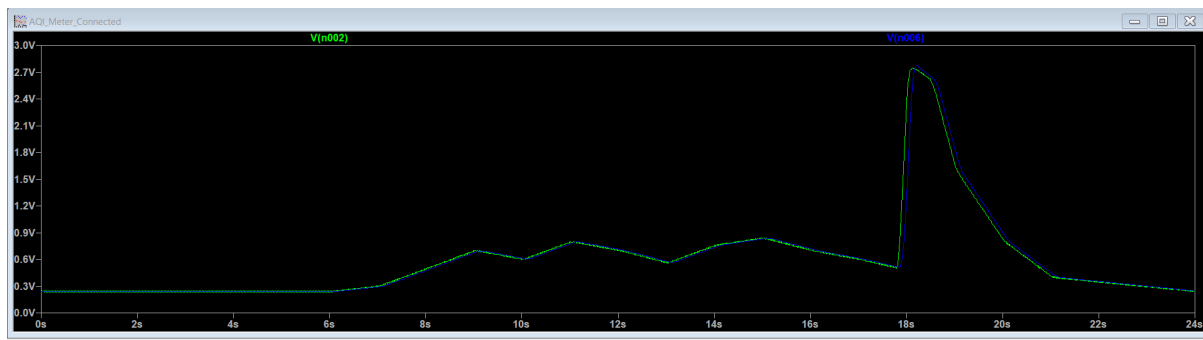


Figure 6: Current converted to voltage for measurement **Green-Initial** -After Filters

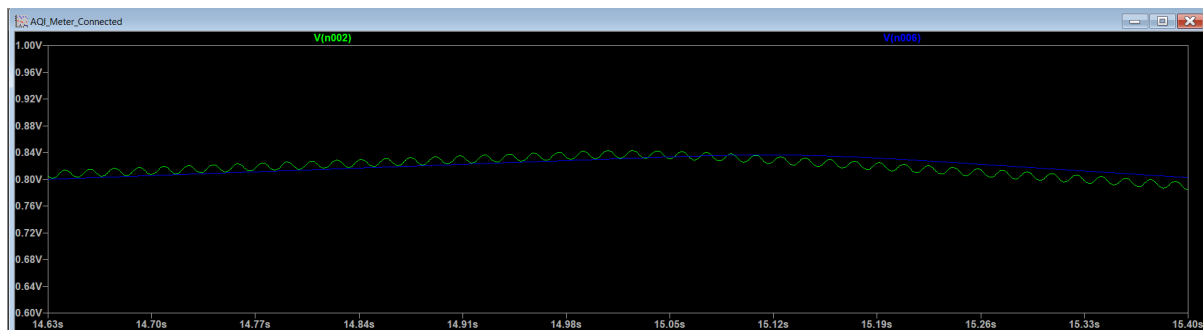


Figure 7: Work of the Filters

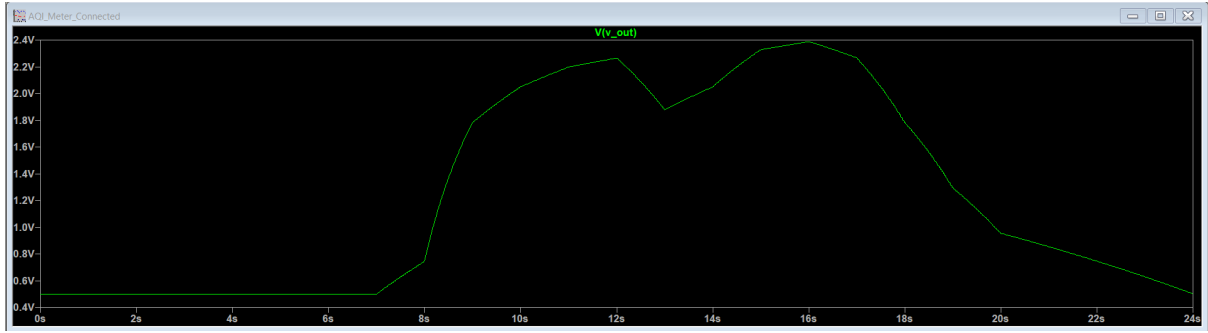


Figure 8: CO2 Mox Sensor Output

5.2 AC Sweep (.ac dec 100 0.1 100k)

- **Purpose:** Frequency-domain validation (Bode Plots).
- **Expected Output:** Verified the -3dB cutoff point of the MFB active filters. Mathematically documented the total dB attenuation precisely at 60Hz to prove EMI rejection. Verified $>60^\circ$ Phase Margin on the TIA to guarantee stability.

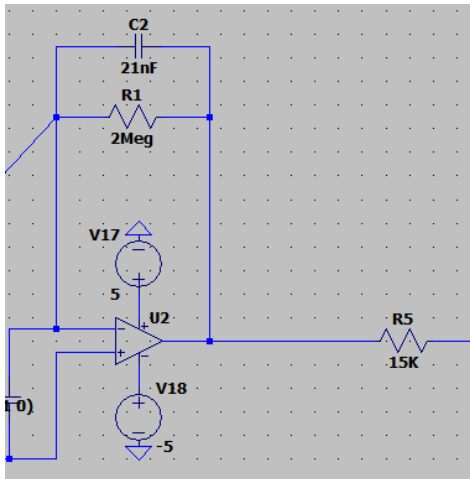


Figure 9: PM2.5 **with** Parasitic Capacitance

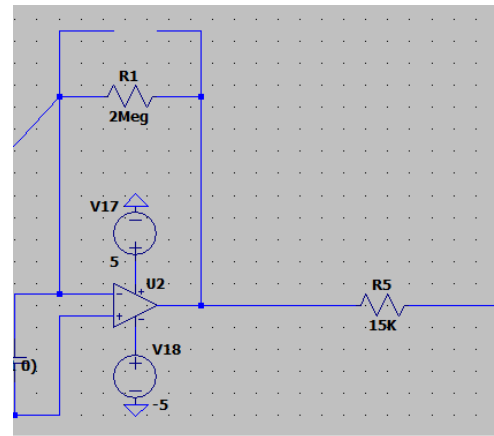


Figure 10: PM2.5 **without** Parasitic Capacitance

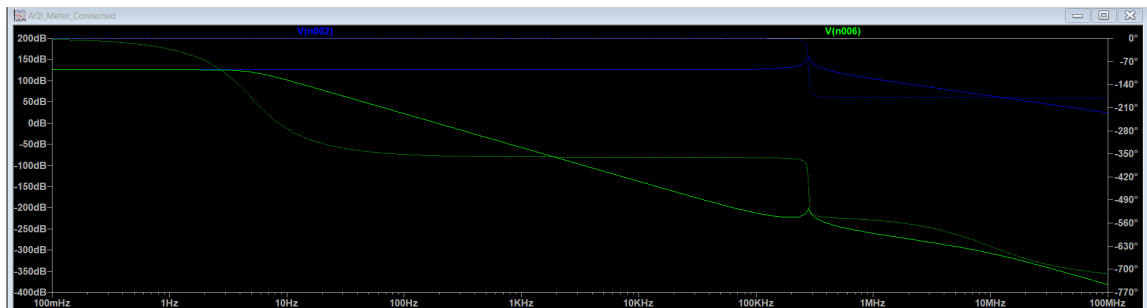


Figure 11: PM2.5 Bode Plot (**Bad** Phase Margin)

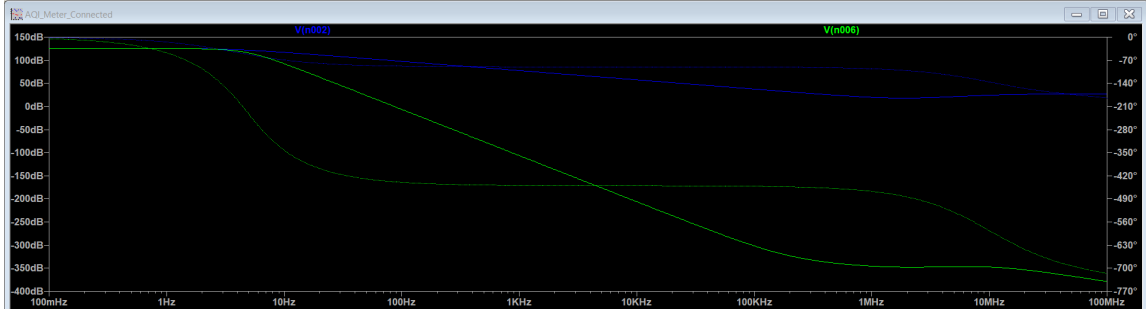


Figure 12: PM2.5 Bode Plot (Good Phase Margin)

5.3 DC Sweep (.dc V_fault -5 5 0.1)

- **Purpose:** Destructive fault validation.
- **Expected Output:** By injecting a sweeping -5V to +5V source into the protection chain, the graph yielded a straight diagonal line that rigidly snapped flat at **-0.3V** and **3.6V**. This provided absolute visual proof that the microcontroller cannot be destroyed by an analog fault.

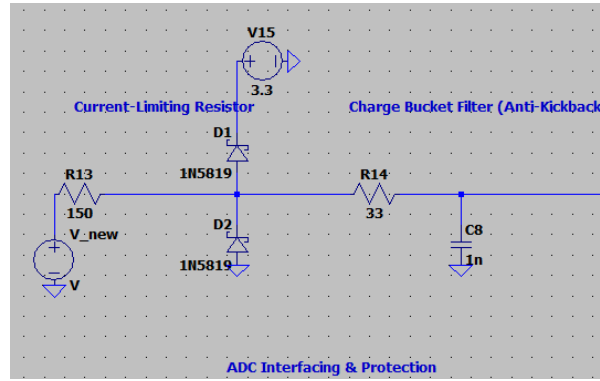


Figure 13: ADC Interfacing -Schottky diodes

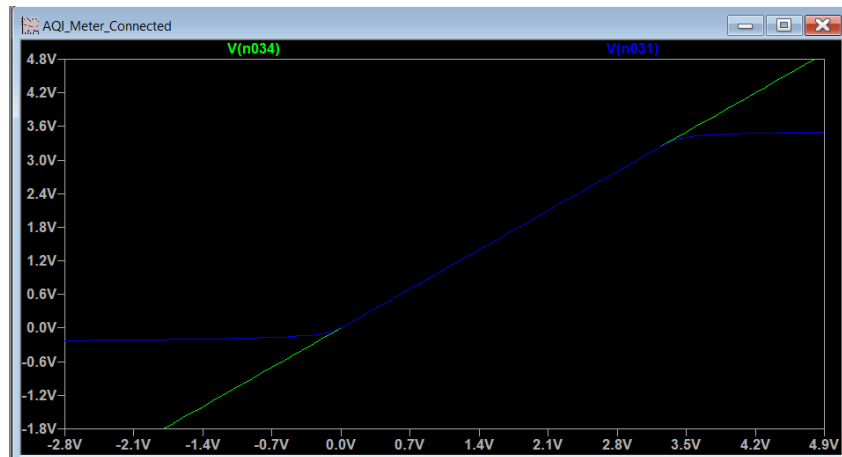


Figure 14: The 3.6V Clamping

5.4 Stability Analysis: The Closed-Loop Gain Peaking Criterion

While exact Phase Margin is traditionally calculated using an open-loop response, the stability of the fully assembled circuit can be accurately inferred directly from the closed-loop AC Bode plot.

The primary indicator of degraded phase margin in a closed-loop system is **Gain Peaking**—a localized amplification spike at the cutoff frequency (the “knee” of the curve) immediately prior to the high-frequency roll-off. System stability is visually evaluated using the following criteria:

- **Marginal Stability (Phase Margin $< 45^\circ$):** Characterized by a sharp upward spike in the gain curve at the cutoff frequency. This peaking indicates severe underdamping, which physically manifests as transient ringing in the time domain and yields a borderline unstable amplifier.
- **Optimal Stability (Phase Margin $> 60^\circ$):** Characterized by a completely flat passband plateau that transitions into a smooth, monotonic roll-off with zero localized peaking. This profile indicates a critically damped or optimally damped system capable of settling quickly without oscillation.